Mechanistic Modelling of the FtsZ-AmpR-AmpC Axis

Calibration against Uehara and Park (2002) and Tests of Generalisability across β-Lactams

Background

- β-Lactamase production is among the most important mechanisms of β-lactam resistance in Gram-negative bacteria (Bush & Bradford, 2020).
- In many Gram-negative bacteria, the chromosomal β-Lactamase gene *ampC*, is inducible and controlled by the transcriptional regulator AmpR (Vadlamani et al., 2015).
- AmpR is a LysR-type transcriptional regulator that can act as either a repressor or an activator of *ampC*, depending on which muropeptide ligand occupies its effector-binding domain (Vadlamani et al., 2015).
- Under normal growth, the cell-wall precursor **UDP-MurNAc-pentapeptide** binds AmpR and represses *ampC*; during β-Lactam stress, **1,6-anhydromuropeptides** accumulate in the cytoplasm and are proposed to displace it, converting AmpR into an activator (Dik et al., 2018).

Motivation (Does Cell Division Influence AmpC Induction?)

- FtsZ, a tubulin-like GTPase, polymerises into the Z-ring at midcell at the onset of division, where it acts as the scaffold for assembly of the divisome (Adams and Errington, 2009).
- In *E. coli* carrying the *C. freundii ampR-ampC* system, β-Lactamase induction requires active FtsZ (Ottolenghi & Ayala, 1991).
- Division mutants accumulate the repressor **UDP-MurNAc-pentapeptide**, which accounts for the loss of AmpC induction in β-Lactam-stressed conditions (Uehara & Park, 2002).
- These findings come from an engineered chimeric system. Species differ in peptidoglycan recycling and cell-wall-stress responses, and how division perturbations translate into AmpC induction in **native** pathogens has not been established (Gupta et al., 2026).

Hypothesis

Building on the *E. coli-C. freundii* system described by Uehara and Park (2002), we hypothesise that FtsZ inactivation in **native *P. aeruginosa*** will suppress β-Lactam-induced AmpC expression by promoting the accumulation of UDP-MurNAc-pentapeptide.

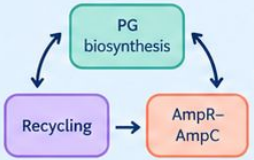
A Note on the Approach

- A kinetic model integrating peptidoglycan biosynthesis, recycling, and AmpC induction was constructed for *P. aeruginosa*, covering most of the corresponding reactions, to test the hypothesis in silico.
- No experimental dataset combining FtsZ perturbation, β-Lactam exposure, and AmpC or muropeptide readouts was found for *P. aeruginosa*, so the model outputs could not be tested.
- The only such dataset is from the *E. coli-C. freundii* chimeric system (Uehara and Park, 2002).
- An analogous model of the chimeric system was therefore built, so that its outputs could be tested directly against the Uehara and Park (2002) measurements.
- The success criteria will be to reproduce the AmpC fold-induction in wild-type and *ftsZ84* cells, and the direction of the repressor (UDP-MurNAc-pentapeptide) changes.
- Agreement would support the modelling approach and justify extending it to *P. aeruginosa*. It would not validate the *P. aeruginosa* model, and its predictions would remain hypothetical pending experiment.

Methodology

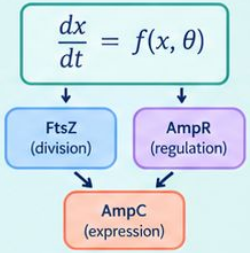
Workflow

1 Pathway Mapping and Integration



- Identified pathways directly relevant to the hypothesis.
- Mapped peptidoglycan biosynthesis, recycling and AmpC regulation.
- Connected pathways through shared metabolites and fluxes.

2 Kinetic ODE Model Construction



- Formulated mass-balance ODEs for all species.
- Integrated kinetics with FtsZ, AmpR and AmpC.
- Implemented the AmpR regulatory switch.
- Added perturbation parameters for the experimental setup.

3 Parameterisation



- Sourced kinetic parameters from literature and BRENDA.
- Estimated/assumed values where data were unavailable.
- Derived values using steady-state assumptions where possible.

4 Initial Model Evaluation



- Simulated with literature-derived and estimated parameters (uncalibrated).
- Compared outputs with experimental AmpC induction and repressor measurements.
- Identified discrepancies with experimental targets.

5 Manual and Automated Calibration



- Manually adjusted key parameters to improve agreement with data.
- Defined a loss function to quantify deviations.
- Applied differential evolution for global optimisation, followed by local refinement.

6 Fit and Parameter Assessment



- Evaluated fit to AmpC fold-induction and UDP-MurNAc-pentapeptide measurements.
- Assessed plausibility of best-fit parameter values.
- Investigated residual mismatches and model structural limitations.

7 Cross-Drug Testing



- Extended simulations to additional β-lactams.
- Kept shared calibrated parameters; added drug-specific properties and kinetics.
- Compared predicted responses with expected or reported AmpC induction behaviour.

Map → Build → Parameterise → Test (uncalibrated) → Calibrate → Assess → Generalise

1. Pathway Mapping and Integration

- Three pathways—**peptidoglycan biosynthesis (map00550)**, **cell wall recycling**, and **AmpC induction** were mapped from the KEGG PATHWAY Database and relevant literature, and combined to get a holistic topological view of the network.
- Cell wall recycling steps were heavily dependent on the extensive review conducted by Dik et al. (2019).
- Modelling of the AmpC induction and deciding on activator identities and binding affinities were built on the foundational work done by the Mobashery lab.

2. Kinetic ODE Model Construction

- Each reaction in the combined network is now mathematically characterised by mass-balance equations of the following form:

$$\frac{d[M]}{dt} = v_{production} - v_{consumption} - \mu[M]$$

  The rate of change of a metabolite's intracellular concentration equals its rate of **production** minus its rate of **consumption** and its rate of **growth-associated dilution**.

- Each production or consumption term is assumed to follow Michaelis-Menten kinetics.

$$v_0 = \frac{V_{max} \cdot [S]}{K_M + [S]}$$

3. Parameterisation

- Kinetic parameters for each rate term, such as **maximum velocity**, **catalytic rate constant**, **Michaelis-Menten constant**, and **equilibrium constant of inhibition** were obtained from the BRENDA database and literature searches.
- Parameters whose values did not exist on any database or literature were guessed or derived from steady-state assumptions.
- The current model comprises of **48** coupled nonlinear ordinary differential equations and **127** parameters. It is a highly underdetermined system (Check this companion module for more details).

4. Initial Model Validation

- The model is specified in Antimony and simulated in Python using Tellurium, a systems biology environment built on libRoadRunner, a high-performance SBML simulation engine. libRoadRunner numerically integrates the ODE system with the CVODE solver and returns the time course of every state variable.
- The timecourse of each metabolite is tracked across **four test conditions** reflecting the experiments performed by Uehara and Park (2002). The four test conditions are:-
  - **WT, no BL (Condition 1)**: Baseline control
  - **WT, +BL (Condition 2)**: Antibiotic only
  - **ftsZ84, no BL (Condition 3)**: ftsZ mutant leading to its depletion/inactivation
  - **ftsZ84, +BL (Condition 4)**: ftsZ mutant with antibiotic (For more details, refer Uehara and Park, 2002)

4. Initial Model Validation (contd.)

#	Target	Definition	Value	What it represents	Weight
1	AmpC fold-induction, WT	$AmpC_2$ / $AmpC_1$	**6.2**	Induction by cefoxitin in wild type (0.5 → 3.1)	1.0
2	AmpC fold-induction, ftsZ84	$AmpC_4$ / $AmpC_3$	**1.4**	Blunted induction when division is blocked (0.5 → 0.7)	1.0
3	AmpC genotype parity, no drug	$AmpC_1$ / $AmpC_3$	**1.0**	No AmpC difference between genotypes without drug	1.0
4	Repressor change with drug, WT	UDP-MurNAc-pentapeptide$_1$ / $_2$	**3.7**	Repressor falls under cefoxitin in wild type	1.0
5	Repressor accumulation, ftsZ84	UDP-MurNAc-pentapeptide$_3$ / $_1$	**3.29**	Division block raises the repressor without drug	1.0
6	Repressor change with drug, ftsZ84	UDP-MurNAc-pentapeptide$_3$ / $_4$	**1.68**	Repressor falls under cefoxitin in ftsZ84	1.0
7	Baseline repressor level (secondary)	UDP-MurNAc-pentapeptide$_1$ (absolute)	**≈ 0.2 mM**	Physiological pool size, as a sanity check	0.5

The model is assessed (and later fitted) against **seven** targets described here. All target values except #7 have been obtained from Uehara and Park, 2002's study (See Table 2 and Figure 3). Target 7 is taken from Mengin-Lecreulx et al. (1982).

More priority is given to meeting the AmpC induction fold-change values as compared to that of the repressor. Hence, the lower weight of target 7.

5. Manual and Automated Calibration

- The uncalibrated model did not reproduce the AmpC induction seen by Uehara and Park (2002).
- Calibration was treated as an *inverse problem*—adjust the unknown parameters until the output matches the data.
- Manual tuning of a few parameters narrowed the gap but could not meet all targets at once.
- Automated calibration followed. **Differential evolution** searched for parameter sets that minimise the gap to the targets.

5. Manual and Automated Calibration (contd.)

$$L(\theta) = \sum_{i=1}^{6} w_i \left(\frac{y_i(\theta)}{\hat{y}_i} - 1 \right)^2 + w_{UDP} \left(\frac{[\text{UDP-NAM-penta}]_1(\theta)}{0.2\ \text{mM}} - 1 \right)^2$$

This is the **loss function** that is minimised to reach the targets using differential evolution. Following are the coefficients and their significance:

- θ: the free (uncertain) parameters being fitted
- $y_i(\theta)$: simulated value of target i; $\hat{y}_i$: its experimental target
- $w_i = 1$ for the six primary targets; $w_{UDP} = 0.5$ for the secondary target
- $[\text{UDP-NAM-penta}]_1$: Repressor concentration in condition 1 (WT, no drug)

6. Fit and Parameter Assessment

- Simulated values are compared with each calibration target at 30 min: AmpC fold-induction (wild-type and *ftsZ84*), genotype parity without drug, and the repressor ratios.
- The final loss is reported alongside a breakdown by target, to show which terms still drive the residual.
- Time courses of AmpC, the activators (*Anh_P*, *Disacch_P_cyto*), and the repressor are plotted for all four conditions. They are checked for direction as well as magnitude: activators and AmpC should rise with drug, and the repressor should accumulate in *ftsZ84*.

7. Cross-Drug Testing

- **Purpose**: To test whether the cefoxitin-calibrated model captures the induction mechanism or only a cefoxitin-specific fit.
- **Method**: To replace only the drug-specific parameters. All cellular parameters (biosynthesis, recycling, AmpR switch, fitted values) stay fixed, and the same four conditions and protocol are used.
- The next slide shows the additional drugs tested on the model.

7. Cross-Drug Testing (contd.)

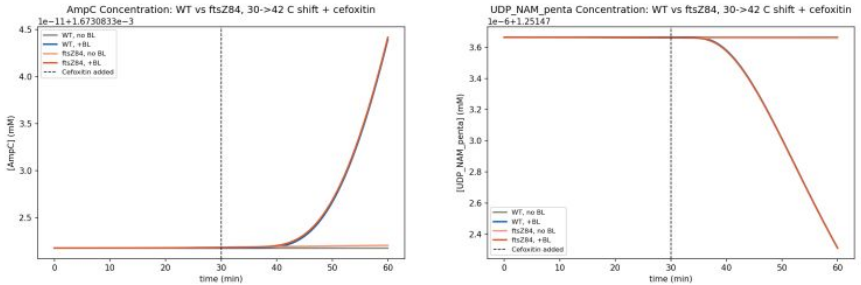
Drug Classification and Properties

Drug	Class	PBP selectivity	AmpC substrate
Ampicillin	Penicillin	PBP4	Good
Cephalexin	Cephalosporin	PBP4	Good
Cefotaxime	Cephalosporin	PBP3	Poor
Cefuroxime	Cephalosporin	PBP3	Poor
Oxacillin	Penicillin	PBP3 and PBP4	Poor

The panel was limited to drugs with AmpC kinetics available for the *C. freundii* enzyme (Bauvois et al., 2005), and it spans **PBP selectivity** and **substrate quality**.

Results

Results of the Uncalibrated Model



(a) AmpC concentration

(b) UMP concentration

(c) Anh_P concentration

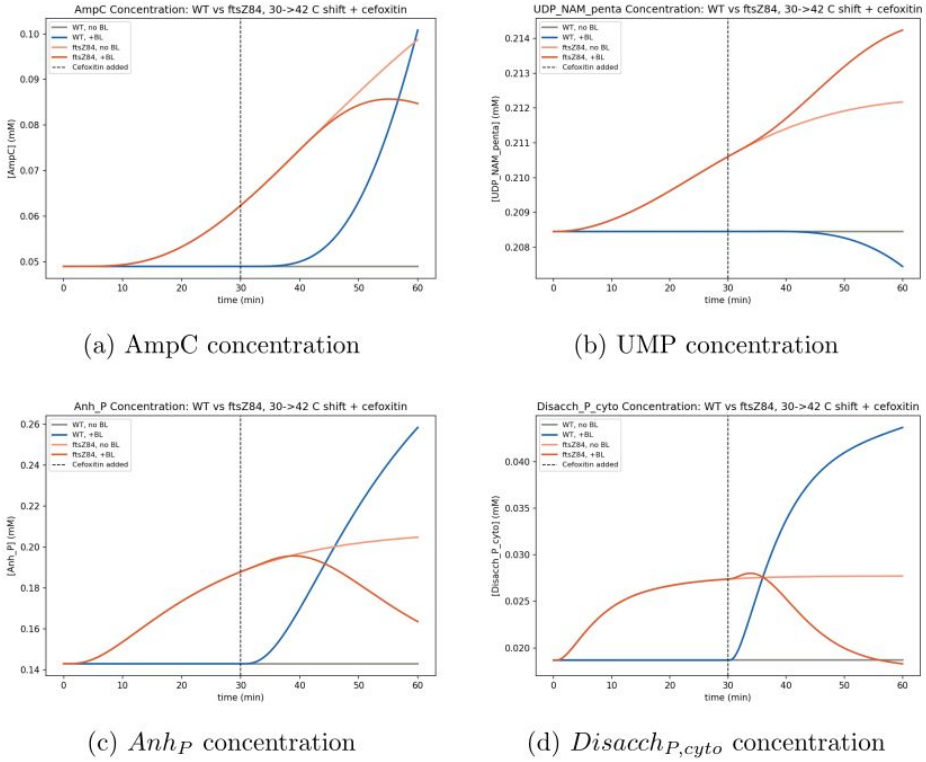
(d) $Disacch_{P,cyto}$ concentration

Figure 1: AmpC and modulator pools in the uncalibrated model

- **AmpC and the repressor (UDP-MurNAc-pentapeptide) are effectively static** in all four conditions. Changes sit at the sixth decimal place or beyond, even after cefoxitin addition, so there is no AmpC induction.
- **The activators do respond to cefoxitin.** Anh_P rises about 66-fold and Disacch_P,cyto about 73-fold, but both stay sub-micromolar, so the change is invisible at the plotted scale.
- **The repressor is stuck at about 1.25 mM**, roughly six times its measured physiological level of about 0.2 mM. The cause is a flux bottleneck in undecaprenyl-phosphate (Und-P) recycling in the uncalibrated parameter set.
- **The activator signal is too small to compete.** Even after a 66-fold rise, Anh_P remains about 10^4-fold below its AmpR binding constant.

Takeaway: The pathway responds in the right direction, but an out-of-range repressor pool alone blocks induction in every condition. This motivates calibration.

Results of the Manually Calibrated Model



(a) AmpC concentration

(b) UMP concentration

(c) Anh_P concentration

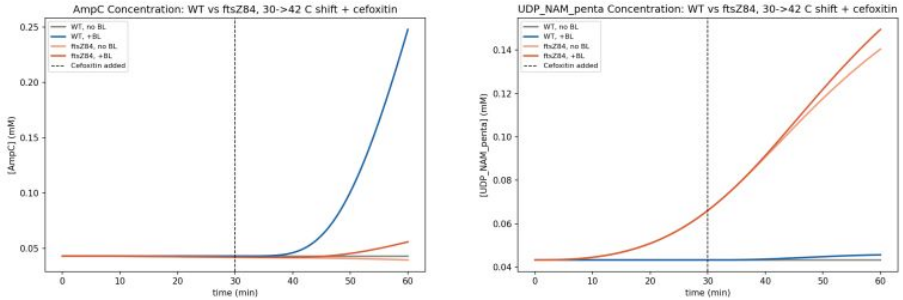
(d) $Disacch_{P,cyto}$ concentration

Figure 2: AmpC and modulator pools in the manually calibrated model

- **WT + cefoxitin (condition 2)**: AmpC rises from 0.049 to 0.101, a 2.06-fold induction. This is a genuine drug-dependent response, but well short of the 6.2-fold target (Figure 2a).
- **ftsZ84 (condition 4 vs 3)**: AmpC falls from 0.099 to 0.085, a fold-change of 0.86 against a target of 1.4. Cefoxitin suppresses AmpC below the no-drug baseline, where the literature shows a smaller but positive induction.
- Condition 4 sits below condition 2, which matches the direction of the hypothesis.
- **Spurious induction in ftsZ84**: AmpC and the activator Anh_P rise before cefoxitin is added (by about 27% and 31% respectively at the time of drug addition). FtsZ inactivation alone is raising the activator signal, independent of drug.

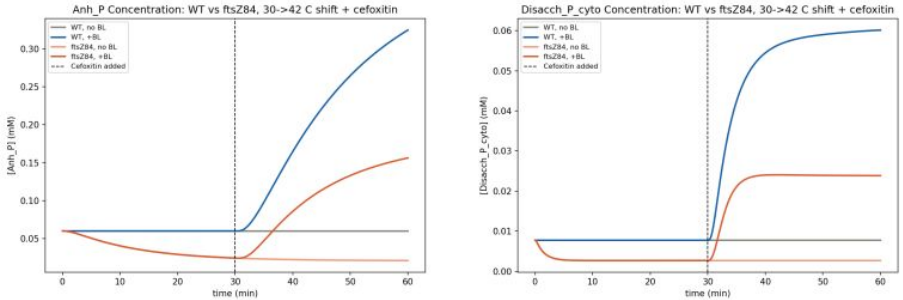
Takeaway: Manual calibration restored induction in wild type but not the *ftsZ84* phenotype, motivating automated calibration.

Results of the Automated Calibrated Model



(a) AmpC concentration

(b) UMP concentration



(c) Anh_P concentration

(d) $Disacch_{P,cyto}$ concentration

Figure 4: AmpC and modulator pools in the automated calibrated model

(a) **AmpC**: Flat at 0.043 mM until cefoxitin is added at 30 min. WT, +BL then shoots up to 0.25 mM (5.8-fold), but *ftsZ84*, +BL barely moves (0.056 mM, about 4-fold lower). No-BL controls stay flat. This is what the hypothesis predicts: no working FtsZ, much less AmpC induction.

(b) **Repressor (UDP-MurNAc-pentapeptide)**: Stays low in WT (0.043 mM) but builds up in both *ftsZ84* conditions, reaching 0.15 mM with BL, about 3-fold above WT, +BL. This supports the hypothesis, since blocked division means less precursor is used up. One catch is that the repressor rises slightly with drug, while the literature reports a drop.

(c) **Anh_P (activator)**: Constant in WT, no BL (0.06 mM) and climbs to 0.32 mM in WT, +BL after cefoxitin. In *ftsZ84*, it falls to 0.02 mM even before the drug and only recovers to 0.16 mM with BL, about half of WT, +BL. Less activator means weaker AmpC induction, as expected.

(d) **Disacch_P_cyto (activator)**: Flat at 0.008 mM in WT, no BL and jumps to 0.06 mM within minutes of cefoxitin in WT, +BL. In *ftsZ84*, it drops to 0.003 mM early on and reaches only 0.024 mM with BL, about 2.5-fold below WT, +BL. This matches panel (c). A smaller activator pool explains the blunted AmpC.

Calibrated Model Reproduces AmpC Fold-Induction Targets

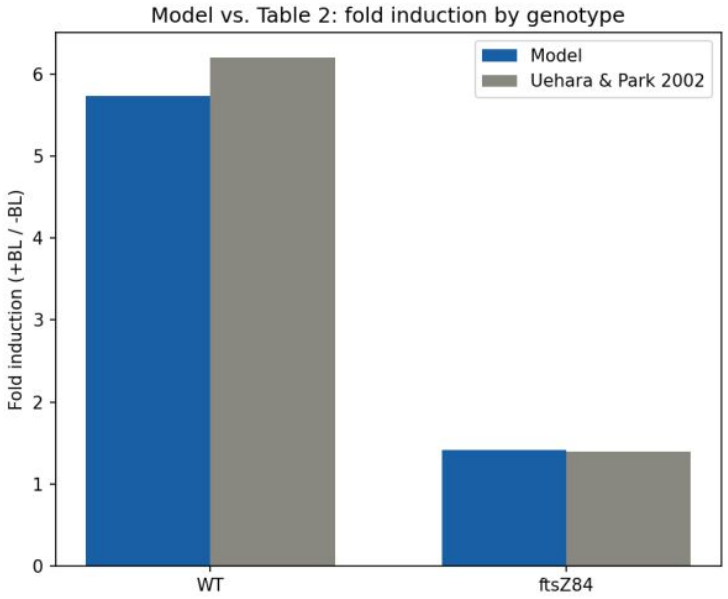
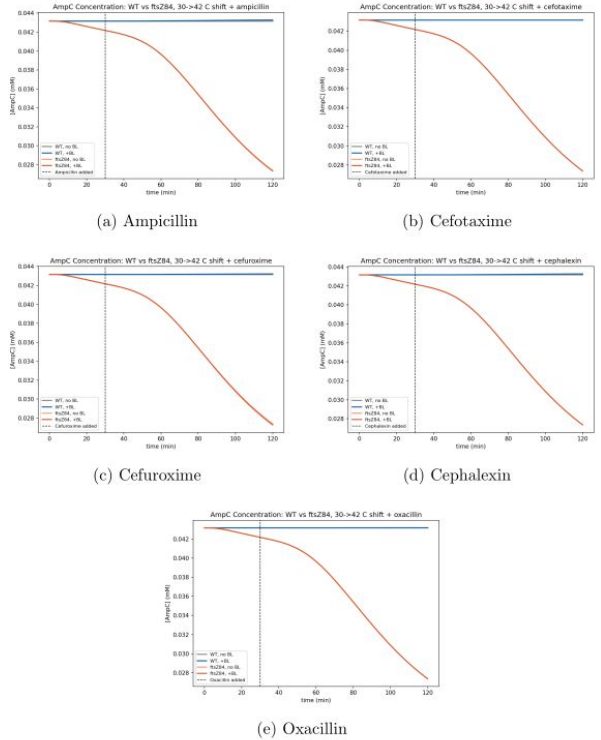


Figure 5: Fold Change Targets (Automated Calibrated Model)

- **WT**: Model predicts 5.7-fold AmpC induction against the literature target of 6.2 (Uehara and Park, 2002), a gap of 8%.
- **ftsZ84**: Model predicts 1.41-fold against the target of 1.40, essentially exact.
- Both genotypes are matched, including the sharp drop in induction when FtsZ is inactive.
- Clear improvement over the manually calibrated model.
- **Caveat**: The repressor still moves opposite to the literature (increase under BL instead of a decrease).

Takeaway: The model reproduces the AmpC induction targets in WT and ftsZ84 cells, but the repressor mismatch marks a gap that must be resolved before extending to *P. aeruginosa*.

Testing on Different Drugs



(a) Ampicillin (b) Cefotaxime

(c) Cefuroxime (d) Cephalexin

(e) Oxacillin

Figure 6: AmpC concentrations, as predicted by the model, under different drugs

- **Same output for every drug**: Ampicillin, cefotaxime, cefuroxime, cephalexin, and oxacillin give near-identical AmpC curves.
- **WT**: AmpC stays flat at 0.043 mM, so no induction is predicted for any drug.
- ***ftsZ84***: AmpC falls by 37% over 120 min (0.043-0.027 mM), regardless of the drug added.
- **Drug identity has no visible effect**: The drugs differ by more than 2 orders of magnitude in their acylation rates and in their main PBP targets (PBP3, PBP4, or multiple PBPs), yet the output is unchanged. The *ftsZ84* term alone appears to drive the decline.
- **Likely cause**: Induction in the model depends mainly on PBP5/6 inhibition. Cefoxitin acylates PBP5/6 efficiently while these five drugs are 10^4-fold weaker there, even though several are potent at PBP3 or PBP4.

Takeaway: Calibration to cefoxitin does not transfer to other β-lactams. Induction depends almost solely on PBP5/6 inhibition

Limitations

- **Underdetermined fit**: 127 parameters against a few targets, so the best-fit values are one consistent solution, not unique estimates.
- **Repressor trend**: The repressor rises slightly with drug where the literature reports a fall, so the AmpC match may not rest on the full mechanism.
- **Poor transfer across drugs**: Little or no induction for the other β-lactams, including ampicillin, a recognised strong inducer. Induction in the model runs mainly through PBP5/6, while the literature points to PBP4 (Sanders et al., 1997).
- **Model simplifications**: Fixed growth rate, lumped enzymes, no CreBC pathway, and FtsZ gating of PBP1b that is probably too strict, which drives a drug-independent AmpC decline in *ftsZ84*.
- Drug parameters come from different strains and are partly estimated.

Future Work

- **Fixing known model issues**: Relaxing the FtsZ gating of PBP1b and adding growth-rate coupling.
- **Resolving the repressor discrepancy**: Testing candidate mechanisms by which β-lactam stress lowers the precursor pool.
- **Adding constraints, then analysing sensitivity**: Running sensitivity analysis once the recalibrated model is stable.
- **Re-testing across drugs**: Re-deriving PBP affinities and compare strong and weak inducers.
- **Extending to P. aeruginosa** with its specific features (CreBC sensing, multiple AmpD paralogues).

Conclusion

Conclusion

- A kinetic model of the FtsZ-AmpR-AmpC axis (peptidoglycan biosynthesis, recycling, and AmpC induction) was built for the *E. coli-C. freundii* system and calibrated to Uehara and Park (2002) using differential evolution.
- The calibrated model reproduces the 30-min AmpC fold-induction: 5.7 against 6.2 in wild-type and 1.41 against 1.40 in ftsZ84. FtsZ inactivation blunts induction, and the repressor builds up in *ftsZ84*, as hypothesised.
- Two gaps remain: the repressor rises slightly with drug where the literature reports a fall, and the model gives little or no induction for other β-lactams, including ampicillin.
- The *P. aeruginosa* extension should wait until these gaps are resolved.

Appendix

For more information on the model architecture and the parameters considered, refer to this companion report.

A GitHub repository has been created for reproducing the results of this study.